

EC466: Econometrics

Problem Set # 8: Callaway-Sant'Anna Estimator and Modern DiD Methods

Background: Consider a training program that was implemented across different cities at different times. You have panel data from five cities (E, F, G, H, I) over seven time periods (2017-2023), with the following average employment rates (in percentage points):

City	2017	2018	2019	2020	2021	2022	2023	Treatment Start
E	62.5	63.2	70.8	74.1	77.5	80.2	82.8	2019
F	61.8	62.4	63.1	71.2	74.9	78.3	81.5	2020
G	62.1	62.9	63.6	64.3	72.8	76.5	79.9	2021
H	63.3	64.0	64.7	65.4	66.1	74.2	77.8	2022
I	61.2	61.8	62.4	63.0	63.6	64.2	64.8	Never

Each city has 150 observations per period, and treatment is irreversible once adopted.

1. Group-Time Average Treatment Effects

- (a) Define $ATT(g, t)$ precisely in the context of this staggered adoption design. What does g represent? What does t represent? What is the interpretation of $ATT(2020, 2022)$ in this application?

- *ANS: $ATT(g, t)$ is the average treatment effect on the treated at time t for the cohort first treated at time g . Here, g represents the treatment adoption year (cohort identifier), and t represents the calendar time period. $ATT(2020, 2022)$ measures the causal effect of the training program in 2022 for City F, which was first treated in 2020—this captures the treatment effect two years post-adoption for this cohort.*

- (b) Using City I (never treated) as the comparison group, calculate $ATT(2019, 2021)$ for City E "by hand" using the unconditional estimand:

$$ATT_{unc}^{nev}(g, t) = E[Y_t - Y_{g-1} | G_g = 1] - E[Y_t - Y_{g-1} | G_g = \infty]$$

Show all your work and interpret the result.

- *ANS: For City E with $g = 2019$, $t = 2021$, $g - 1 = 2018$:
Treated group (City E): $Y_{2021} - Y_{2018} = 77.5 - 63.2 = 14.3$
Never-treated (City I): $Y_{2021} - Y_{2018} = 63.6 - 61.8 = 1.8$
Therefore: $ATT(2019, 2021) = 14.3 - 1.8 = 12.5$
Interpretation: The training program increased employment rates by 12.5 percentage points in 2021 for City E, two years after treatment adoption, relative to the counterfactual trend.*
- (c) Now calculate $ATT(2020, 2022)$ for City F using the same approach. How does this estimate compare to the one you calculated in part (b)? What might explain the difference?
- *ANS: For City F with $g = 2020$, $t = 2022$, $g - 1 = 2019$:
Treated group (City F): $Y_{2022} - Y_{2019} = 78.3 - 63.1 = 15.2$
Never-treated (City I): $Y_{2022} - Y_{2019} = 64.2 - 62.4 = 1.8$
Therefore: $ATT(2020, 2022) = 15.2 - 1.8 = 13.4$
This estimate (13.4) is larger than part (b)'s estimate (12.5). Possible explanations include: treatment effect heterogeneity across cohorts, with City F experiencing stronger effects; cumulative or growing treatment impacts over time; or differential baseline characteristics between cohorts affecting treatment response.*

2. Alternative Comparison Groups

- (a) Explain the difference between using “never-treated” units versus “not-yet-treated” units as a comparison group. Write down the parallel trends assumption for each case.
- *ANS: Never-treated units are those that never receive treatment during the observed period (City I), while not-yet-treated units are those that will eventually receive treatment but have not been treated by time t . The parallel trends assumption for never-treated: $E[Y_t(0) - Y_{g-1}(0)|G_g = g] = E[Y_t(0) - Y_{g-1}(0)|G_g = \infty]$, meaning the treated cohort would have followed the same trend as never-treated units absent treatment. For not-yet-treated: $E[Y_t(0) - Y_{g-1}(0)|G_g = g] = E[Y_t(0) - Y_{g-1}(0)|G_g = g']$ where $g' > t$, meaning the treated cohort would have followed the same trend as cohorts not yet treated by time t . Not-yet-treated units may be more comparable if treatment timing reflects similar underlying characteristics, but may exhibit anticipation effects.*

- (b) Using City H as a “not-yet-treated” comparison group, calculate $ATT(2020, 2021)$ for City F. How does this estimate differ from using City I (never-treated) as the comparison? Which comparison do you find more credible and why?

- *ANS: For City F with $g = 2020$, $t = 2021$, using City H (treated in 2022, so not-yet-treated in 2021):*

$$\text{Treated (City F): } Y_{2021} - Y_{2019} = 74.9 - 63.1 = 11.8$$

$$\text{Not-yet-treated (City H): } Y_{2021} - Y_{2019} = 66.1 - 64.7 = 1.4$$

$$ATT(2020, 2021) = 11.8 - 1.4 = 10.4$$

$$\text{Using City I: } Y_{2021} - Y_{2019} = 63.6 - 62.4 = 1.2, \text{ giving } ATT(2020, 2021) = 11.8 - 1.2 = 10.6$$

The estimates are similar (10.4 vs 10.6). City H may be more credible if cities that adopt treatment share common pre-treatment trends, making not-yet-treated units better counterfactuals. However, City H could exhibit anticipation effects. City I is credible if it represents a stable counterfactual, but may differ systematically if non-adoption reflects structural differences.

- (c) Suppose you wanted to estimate $ATT(2019, 2023)$ for City E. Which cities could serve as valid “not-yet-treated” comparison groups in 2023? Explain your reasoning. What does this imply about the availability of comparison groups over time?

- *ANS: No cities can serve as not-yet-treated comparison groups in 2023. By 2023, all cities except I have been treated: F (2020), G (2021), and H (2022) are all already treated. Only City I remains, but it is never-treated, not not-yet-treated. This illustrates a fundamental challenge in staggered adoption designs: the pool of not-yet-treated comparison units shrinks over time as more cohorts adopt treatment. Later time periods have fewer valid comparison units, potentially reducing precision and making parallel trends less credible for long-run effects.*

3. Aggregation and Summary Measures

- (a) Consider the simple average aggregation:

$$\theta_M := \frac{2}{T(T-1)} \sum_{g=2}^T \sum_{t=2}^T 1\{g \leq t\} ATT(g, t)$$

Calculate the number of group-time ATTs that would be averaged in θ_M

for the data above (i.e., how many (g, t) pairs satisfy $g \leq t$?). Explain why this aggregation might “overweight” earlier-treated cities.

- *ANS: There are 14 (g, t) pairs satisfying $g \leq t$: City E (2019) contributes 5 pairs for $t \in \{2019, 2020, 2021, 2022, 2023\}$; City F (2020) contributes 4 pairs; City G (2021) contributes 3 pairs; City H (2022) contributes 2 pairs. Total: $5 + 4 + 3 + 2 = 14$. This aggregation overweights earlier-treated cities because they appear in more (g, t) pairs—City E appears 5 times while City H appears only 2 times. Each city receives equal weight per pair, so earlier adopters disproportionately influence θ_M , potentially biasing the overall average if treatment effects vary across cohorts or event times.*

(b) Write down the formula for the event-study aggregation $\theta_D(e)$, where e represents periods of exposure to treatment. What is the interpretation of $\theta_D(2)$ in this application?

- *ANS: The event-study aggregation is $\theta_D(e) = \sum_g w_g(e) \cdot ATT(g, g + e)$, where e is event time (periods since treatment), $w_g(e)$ are weights satisfying $\sum_g w_g(e) = 1$, and the sum is over all cohorts g for which $ATT(g, g + e)$ is defined. $\theta_D(2)$ represents the average treatment effect two periods after adoption, averaged across all treatment cohorts. In this application, $\theta_D(2)$ aggregates the effect of the training program two years post-implementation across Cities E, F, G, and H (where applicable), providing a summary measure of the medium-run program impact.*

(c) Compare $\theta_D(2) - \theta_D(1)$ to the difference in individual group effects $ATT(g, g + 2) - ATT(g, g + 1)$ for a specific group g . Under what conditions would these two measures of dynamic treatment effects be identical? When might they differ?

- *ANS: $\theta_D(2) - \theta_D(1)$ measures the change in average treatment effects from event time 1 to event time 2 across all cohorts, while $ATT(g, g + 2) - ATT(g, g + 1)$ measures this change for a specific cohort g . These would be identical if: (1) treatment effects are homogeneous across cohorts (i.e., all groups have the same dynamic treatment trajectory), or (2) only one cohort contributes to both $\theta_D(1)$ and $\theta_D(2)$. They differ when there is treatment effect heterogeneity across cohorts—different cities may experience different growth rates in treatment effects over time—or when the weighting differs across event times due to varying*

cohort composition or sample sizes at different exposure lengths.